

CHARGE-DEPENDENT INSERTION OF THE ANTIMICROBIAL PEPTIDE MAGAININ-2 INTO LIPID MONOLAYERS

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ABSTRACT

Many antimicrobial peptides act by disrupting cellular membranes, with therapeutic selectivity arising from their ability to discriminate between bacterial and mammalian cells, a distinction largely governed by membrane surface charge. Here, we investigate the interfacial organisation and membrane insertion of the cationic antimicrobial peptide magainin 2 using specular neutron reflectometry on lipid monolayers that mimic bacterial and mammalian membranes. Negatively charged dipalmitoylphosphatidylglycerol (DPPG) and zwitterionic dipalmitoylphosphatidylcholine (DPPC) were used as respective representative model systems. Experimental conditions were adjusted to compensate for the markedly different membrane association constants of magainin 2, enabling direct comparison of peptide behavior at the two interfaces.

Despite similar peptide coverages, magainin 2 exhibits strikingly different interfacial behavior. At the DPPG interface, the magainin 2 α -helix inserts deeply and adopts an orientation parallel to the membrane surface, with hydrophobic residues facing the lipid chains and cationic residues interacting with the polar headgroups (Fig. 1A). In contrast, at the DPPC interface, the peptide displays limited interaction with the monolayer and a diffuse intensity distribution

across the headgroup and acyl-chain regions, consistent with a largely disordered conformation (Fig. 1B).

These results demonstrate that electrostatic interactions not only promote membrane binding but also dictate peptide insertion depth and interfacial organisation. By directly linking membrane charge to peptide penetration and structural arrangement, this work provides mechanistic insight into the selective antibacterial activity of magainin 2 and establishes a general framework for understanding the membrane-disruptive action of cationic antimicrobial peptides.

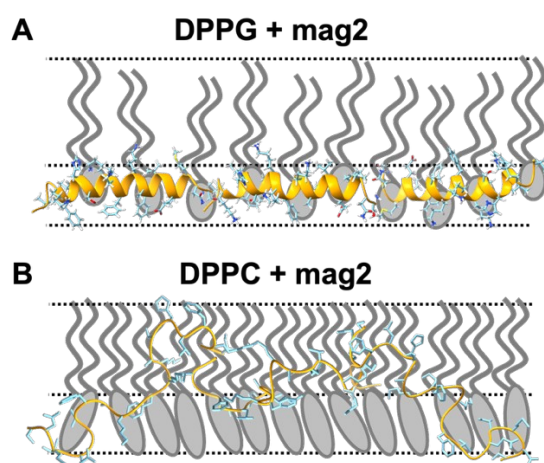


Fig. 1. Localisation of magainin 2 within DPPG and DPPC monolayers at the air-water interface, as determined by neutron reflectivity. The peptide is shown in an α -helical conformation in DPPG (A) and an unstructured conformation in DPPC (B), based on previous solid-state NMR studies (1-3). The peptide backbone is shown in orange, with the side chains shown in cyan.

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